

Q.22 Describe the function and element of robotic vision system. (CO1)

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SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain Robotic subsystem-motion, recognition and control. (CO2)

Q.24 Describe the functions and classifications of robotic sensors in detail. (CO1)

Q.25 Explain the term SCARA and Cartesian configuration of robots. (CO3)

3rd Sem / Automation and Robotics

Subject : Robotics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which mode of robot operation involves movement along x,y, and z axes? (CO5)

- a) Joint mode b) Teach pendant mode
- c) Cartesian mode d) Polar mode

Q.2 Driver are also known as (CO3)

- a) Controller b) Sensor
- c) Manipulator d) Actuator

Q.3 Which of the following make use of robots. (CO1)

- a) Assembly b) welding
- c) Painting d) All of the above

Q.4 Which of the following is an example of an electric actuator? (CO3)

- a) Hydraulic cylinder b) Pneumatic valve
c) Solenoid d) Piston pump

Q.5 Which sensor is used to measure force based on the deformation of a material? (CO4)

- a) Encoder b) Strain gauge
c) Potentiometer d) LVDT

Q.6 What is the full form of DOF? (CO2)

- a) Degree of finance b) Degree of Freedom
c) Degree of Fail d) None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Electric actuators are controlled using techniques like pulse width modulation and _____ control. (CO1)

Q.8 Vision systems in robotics are primarily used for speed control. (True/False) (CO2)

Q.9 Give one example of external sensor. (CO3)

Q.10 Teach pendants have modes like keypad layout, teach mode, and _____ mode. (CO5)

Q.11 Define CAD. (CO1)

Q.12 Define LVDT. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Describe the Laws of Robotics. (CO1)

Q.14 Explain non-industrial robots in brief. (CO2)

Q.15 What are Grippers? Explain mechanical Grippers. (CO3)

Q.16 Explain the concept of kinematic chain. (CO5)

Q.17 Explain Fluid power Actuator control system. (CO5)

Q.18 Define Actuator. Explain Electric Actuators. (CO3)

Q.19 Explain the working of DC electric motor. (CO5)

Q.20 Explain teach pendant. (CO1)

Q.21 Explain the working of joint mode operation in robotics. (CO4)